

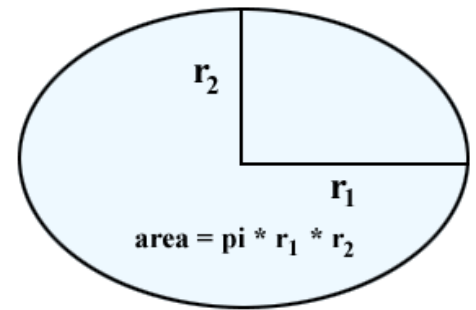
CS 170 Proficiency Quiz 4

Write a program that that accepts the **input**, performs the **processing** and prints the **output** shown below. You have 45 minutes to complete the problem. You may not access any notes (other than the Java Quick Reference below), files, or communicate with anyone in any way (including checking your email or looking up files on the Q: drive.) When you have finished and submitted the quiz, check to see that I have received it.

You can check your problem by using **Tools->Compile** to see if your code is syntactically correct. To **test your code**, use **Tools->Go**. You can **run** the program (in the console) by using **Tools->Run in Console**.

Input, Output and Processing

Create a console program named **EllipseAreaCalc** that calculates the area of an ellipse, given its **width** and **height**. The formula is shown to the right. Remember that you'll need to divide the **width** and **height** by **2** to get the *major* and *minor* axes, **r1** and **r2** used in the illustration. Here is what the output of your program should look like:



```
sgilbert: Ellipse Area Calculator
=====

Enter the width of the ellipse: 2.52
Enter the height of the ellipse: 17.1

Ellipse Area: 33.84438
```

- ✓ Line # 1 must begin with **your** student ID. (Mine is shown above)
- ✓ Line #2 is the same length as Line #1 and uses equals-signed.
- ✓ Lines #3 and #6 are blank
- ✓ Lines #4 and #5 begin with a prompt. Input should occur on the same line.
- ✓ Line #7 must contain the label "**Ellipse Area:** " and end with a newline.
- ✓ The printed area **should not** be formatted. Just use concatenation.
- ✓ Use **Math.PI** for the value of **pi**, not **3.14159**